

45.3 Inference for quantile regression

The estimator of [Section 45.2](#) minimizes a criterion with no derivative at the kink and no closed form, yet its large-sample behaviour is as clean as that of least squares, with one new ingredient: the asymptotic variance involves the *density* of the response at the fitted quantile, which is hard to estimate well. Everything practical below is an attempt either to estimate that density or to avoid it.

45.3.1 An identity of Knight's

The analysis rests on an exact decomposition of the increment of the check loss into a linear part and a controllable remainder.

Lemma 45.3.1. Knight's identity

For all real u and v , with $\psi_\tau(u) = \tau - \mathbf{1}\{u < 0\}$,

$$\rho_\tau(u - v) - \rho_\tau(u) = -v \psi_\tau(u) + \int_0^v \{\mathbf{1}\{u \leq s\} - \mathbf{1}\{u \leq 0\}\} ds. \quad (45.3.1)$$

Proof. Both sides vanish at $v = 0$, so it suffices to check four cases. Suppose first $u > 0$. If $0 < v \leq u$ the left side is $\tau(u - v) - \tau u = -\tau v$ and the integrand vanishes for $s \leq v < u$, so the right side is $-\tau v$ too. If $v > u$ the left side is $(\tau - 1)(u - v) - \tau u = -u - \tau v + v$, while the integral contributes $v - u$, giving $-\tau v + (v - u)$: the same. If $v < 0$ the integral runs backwards over $s \in (v, 0)$, where $s < 0 < u$ makes the integrand zero, and the left side is $-\tau v$ again.

Now suppose $u \leq 0$, so $\psi_\tau(u) = \tau - 1$ and $\mathbf{1}\{u \leq 0\} = 1$. If $v > 0$ then for $s \in (0, v)$ we have $u \leq 0 < s$, the integrand is zero, and both sides equal $-v(\tau - 1)$. If $u \leq v < 0$ the same computation holds with the integral over $(v, 0)$, where $u \leq v < s$ again makes the integrand vanish. If $v < u \leq 0$, the left side is $\tau(u - v) - u(\tau - 1) = u - \tau v$; the integrand equals -1 on (v, u) and 0 on $(u, 0)$, so the backwards integral contributes $u - v$, and the right side is $-v(\tau - 1) + (u - v) = u - \tau v$. ■

45.3.2 The limiting distribution

Theorem 45.3.2. Asymptotic normality of a regression quantile

Fix τ and suppose the rows $\mathbf{x}_{(i)}$ are fixed and the errors $u_i = y_i - \mathbf{x}_{(i)}^\top \boldsymbol{\beta}_\tau$ are independent with distribution functions F_i satisfying $F_i(0) = \tau$. Assume

- (C1) each F_i has a density f_i in a neighbourhood of zero with $f_i(0) > 0$; the f_i are equicontinuous at zero, that is $\sup_i \sup_{|s| \leq t} |f_i(s) - f_i(0)| \rightarrow 0$ as $t \downarrow 0$; and $\sup_i \sup_{|s| < \epsilon} f_i(s) < \infty$ for some $\epsilon > 0$;
- (C2) $n^{-1} \sum_i \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top \rightarrow \mathbf{D}_0$ and $n^{-1} \sum_i f_i(0) \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top \rightarrow \mathbf{D}_1$, both positive definite;
- (C3) $n^{-1/2} \max_i \|\mathbf{x}_{(i)}\| \rightarrow 0$.

Then

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_\tau - \boldsymbol{\beta}_\tau) \xrightarrow{d} N_p(\mathbf{0}, \tau(1-\tau) \mathbf{D}_1^{-1} \mathbf{D}_0 \mathbf{D}_1^{-1}). \quad (45.3.2)$$

If the errors are identically distributed with density f , then $\mathbf{D}_1 = f(0)\mathbf{D}_0$ and the covariance collapses to

$$\frac{\tau(1-\tau)}{f(0)^2} \mathbf{D}_0^{-1} = \tau(1-\tau) s(\tau)^2 \mathbf{D}_0^{-1}, \quad s(\tau) = \frac{1}{f\{F^{-1}(\tau)\}}, \quad (45.3.3)$$

where s is the **sparsity function**, the derivative of the quantile function.

Proof. Write $\boldsymbol{\delta} \in \mathbb{R}^p$, $c_i = \mathbf{x}_{(i)}^\top \boldsymbol{\delta} / \sqrt{n}$ and

$$Z_n(\boldsymbol{\delta}) = \sum_{i=1}^n \{\rho_\tau(u_i - c_i) - \rho_\tau(u_i)\},$$

a convex function of $\boldsymbol{\delta}$ minimized at $\hat{\boldsymbol{\delta}}_n = \sqrt{n}(\hat{\boldsymbol{\beta}}_\tau - \boldsymbol{\beta}_\tau)$. By Eq. 45.3.1,

$$\begin{aligned} Z_n(\boldsymbol{\delta}) &= -\boldsymbol{\delta}^\top \mathbf{w}_n + R_n(\boldsymbol{\delta}), \quad \mathbf{w}_n = \frac{1}{\sqrt{n}} \sum_i \mathbf{x}_{(i)} \psi_\tau(u_i), \\ R_n(\boldsymbol{\delta}) &= \sum_i \int_0^{c_i} \{\mathbf{1}\{u_i \leq s\} - \mathbf{1}\{u_i \leq 0\}\} ds. \end{aligned}$$

Because F_i is continuous at zero and $F_i(0) = \tau$, the variables $\psi_\tau(u_i)$ are independent with mean zero and variance $\tau(1-\tau)$, and they are bounded. Condition (C3) makes the Lindeberg condition hold for $\mathbf{w}_n$, which by (C2) and the Cramér–Wold device gives $\mathbf{w}_n \xrightarrow{d} N_p\{\mathbf{0}, \tau(1-\tau)\mathbf{D}_0\}$.

For the remainder, $E R_n(\boldsymbol{\delta}) = \sum_i \int_0^{c_i} \{F_i(s) - F_i(0)\} ds$. By the equicontinuity in (C1), $F_i(s) - F_i(0) = f_i(0)s + o(s)$ uniformly in i , and $\max_i |c_i| \leq n^{-1/2} \|\boldsymbol{\delta}\| \max_i \|\mathbf{x}_{(i)}\| \rightarrow 0$ by (C3), so

$$\begin{aligned} E R_n(\boldsymbol{\delta}) &= \sum_i \left\{ \frac{1}{2} f_i(0) c_i^2 + o(c_i^2) \right\} \\ &= \frac{1}{2} \boldsymbol{\delta}^\top \left\{ \frac{1}{n} \sum_i f_i(0) \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top \right\} \boldsymbol{\delta} + o(1) \longrightarrow \frac{1}{2} \boldsymbol{\delta}^\top \mathbf{D}_1 \boldsymbol{\delta}. \end{aligned}$$

Each summand of R_n is bounded in absolute value by $|c_i| \mathbf{1}\{|u_i| \leq |c_i|\}$, whose second moment is at most $c_i^2 \cdot 2|c_i| \sup f_i$ once $|c_i| < \epsilon$. Summing the independent terms,

$$\text{Var } R_n(\boldsymbol{\delta}) \leq 2 \left(\sup_i \sup_{|s| < \epsilon} f_i(s) \right) \max_i |c_i| \sum_i c_i^2 \longrightarrow 0,$$

because $\sum_i c_i^2 \rightarrow \boldsymbol{\delta}^\top \mathbf{D}_0 \boldsymbol{\delta}$ and $\max_i |c_i| \rightarrow 0$. Hence $R_n(\boldsymbol{\delta}) \rightarrow \frac{1}{2} \boldsymbol{\delta}^\top \mathbf{D}_1 \boldsymbol{\delta}$ in probability, and for each fixed $\boldsymbol{\delta}$,

$$\begin{aligned} Z_n(\boldsymbol{\delta}) &\xrightarrow{d} Z(\boldsymbol{\delta}) = -\boldsymbol{\delta}^\top \mathbf{w} + \frac{1}{2} \boldsymbol{\delta}^\top \mathbf{D}_1 \boldsymbol{\delta}, \\ \mathbf{w} &\sim N_p\{\mathbf{0}, \tau(1-\tau)\mathbf{D}_0\}, \end{aligned}$$

jointly over finitely many $\boldsymbol{\delta}$, since $\mathbf{w}_n$ is the only random part in the limit. The limit Z is a strictly convex quadratic with unique minimizer $\mathbf{D}_1^{-1} \mathbf{w}$.

The last step is the one this book takes on trust. Each Z_n is convex, so pointwise convergence in distribution to a limit with a unique minimizer implies convergence in distribution of the minimizers: this is the convexity lemma of Pollard (1991), proved there from the fact that convex functions converging pointwise on a dense set converge uniformly on compacta. Applying it, $\hat{\delta}_n \xrightarrow{d} \mathbf{D}_1^{-1} \mathbf{w}$, whose covariance is [Eq. 45.3.2](#). The identically distributed case follows by substituting $f_i = f$ and noting that $s(\tau) = 1/f\{F^{-1}(\tau)\}$ is the derivative of F^{-1} at τ . ■

Warning (What the sparsity function does to precision). In [Eq. 45.3.3](#) the factor $\tau(1 - \tau)$ shrinks as τ approaches 0 or 1, suggesting that extreme quantiles are estimated *more* precisely. They are not: $s(\tau)^2$ blows up faster, because the tail density is small. For a standard normal $\tau(1 - \tau)s(\tau)^2$ is $0.25/\varphi(0)^2 \approx 1.571$ at the median and about 13.9 at $\tau = 0.99$. Extreme conditional quantiles need far more data, or a model that borrows strength across levels.

45.3.3 Estimating the sparsity function

The iid formula [Eq. 45.3.3](#) requires $f\{F^{-1}(\tau)\}$, a density at a point. The standard device replaces it by a difference quotient of the empirical quantile function of the residuals,

$$\hat{s}(\tau) = \frac{\hat{F}_n^{-1}(\tau + h_n) - \hat{F}_n^{-1}(\tau - h_n)}{2h_n}, \quad (45.3.4)$$

with a bandwidth $h_n \rightarrow 0$ slowly; Hall and Sheather (1988) derived the choice $h_n = n^{-1/3} z_\alpha^{2/3} \{1.5\varphi(z_\tau)^2 / (2z_\tau^2 + 1)\}^{1/3}$ from an Edgeworth expansion for studentized quantiles. The non-identically-distributed case needs $\mathbf{D}_1$, estimated by replacing $f_i(0)$ with a kernel estimate of the density of the i th residual at zero, in the form Koenker (2005, chapter 4) attributes to Powell. Both involve a bandwidth and both converge slowly, which is the reason for the alternatives below.

Example. Three standard errors for one slope

At $\tau = 0.5$ the fitted income slope is 0.5602, and the three routes give very different precisions:

method	$\tau = 0.5$	$\tau = 0.9$
iid formula Eq. 45.3.3	0.0121	0.0143
sandwich Eq. 45.3.2	0.0308	0.0300
pairs bootstrap, $B = 999$	0.0344	0.0264

At the median the iid formula is optimistic by a factor of 2.5 against the sandwich and rather more against the bootstrap, which is no surprise: Example (Engel's budgets, one more time) showed that these errors are anything but identically distributed. The least squares slope has standard error 0.0144 under the same false assumption. The estimated sparsity is 192.2 francs at $\tau = 0.5$ and 378.9 at $\tau = 0.9$.

```

import numpy as np
import statsmodels.api as sm
from scipy import stats
from scipy.optimize import linprog
data = sm.datasets.engel.load_pandas().data

def qreg(X, y, tau):
    """Quantile regression by linear programming (as in Section 45.2)."""
    n, p = X.shape
    cost = np.concatenate([np.zeros(2 * p), tau * np.ones(n), (1 - tau) * np.ones(n)
    ])
    A = np.hstack([X, -X, np.eye(n), -np.eye(n)])
    out = linprog(cost, A_eq=A, b_eq=y, bounds=(0, None), method="highs")
    return out.x[:p] - out.x[p:2 * p]

income = data["income"].to_numpy()
food = data["foodexp"].to_numpy()
n = len(food)
X = np.column_stack([np.ones(n), income])

def hall_sheather(n, tau, alpha=0.05):
    """Bandwidth in the tau direction for a difference quotient of the quantile
    function."""
    z, za = stats.norm.ppf(tau), stats.norm.ppf(1 - alpha / 2)
    return n ** (-1 / 3) * za ** (2 / 3) * (1.5 * stats.norm.pdf(z) ** 2 / (2 * z **
    2 + 1)) ** (1 / 3)

def sparsity(resid, tau, n):
    """Siddiqui's estimate of  $s(\tau) = 1 / f(F^{-1}(\tau))$ : a difference quotient."""
    h = min(hall_sheather(n, tau), tau, 1 - tau)
    hi, lo = np.quantile(resid, [min(tau + h, 1.0), max(tau - h, 0.0)])
    return (hi - lo) / (2 * h)

def se_iid(X, y, tau):
    """Standard errors under independent and identically distributed errors."""
    beta = qreg(X, y, tau)
    s = sparsity(y - X @ beta, tau, len(y))
    return beta, np.sqrt(tau * (1 - tau) * s ** 2 * np.diag(np.linalg.inv(X.T @ X)))

def se_sandwich(X, y, tau):
    """Powell's sandwich: a kernel estimate of the matrix  $\sum_i f_i(0) x_i x_i'$ ."""
    beta = qreg(X, y, tau)
    n = len(y)
    r = y - X @ beta
    h = sparsity(r, tau, n) * hall_sheather(n, tau) # a window on the residual
    scale
    dens = np.exp(-0.5 * (r / h) ** 2) / (h * np.sqrt(2 * np.pi))
    D1 = np.linalg.inv(X.T @ (dens[:, None] * X))
    return beta, np.sqrt(tau * (1 - tau) * np.diag(D1 @ (X.T @ X) @ D1))

def se_bootstrap(X, y, tau, B, rng):
    """The pairs bootstrap of chapter 23: resample  $(y_i, x_i)$  with replacement and
    refit."""
    draws = np.array([qreg(X[i], y[i], tau)
    for i in rng.integers(0, len(y), (B, len(y)))]])
    return draws.std(axis=0, ddof=1)

```

```

beta50, se50_iid = se_iid(X, food, 0.50)
_, se50_sand = se_sandwich(X, food, 0.50)
se50_small = se_bootstrap(X, food, 0.50, 199, np.random.default_rng(4545))
print(f"slope at tau = 0.5: {beta50[1]:.4f}")
print(f"  se: iid errors {se50_iid[1]:.4f}, sandwich {se50_sand[1]:.4f}, "
      f"  bootstrap with B = 199 {se50_small[1]:.4f}")

```

[Run this code online](#)

45.3.4 Avoiding the density

Two families of methods sidestep [Eq. 45.3.4](#) altogether.

Rank-score tests. The dual variables of [Eq. 45.2.4](#), described in Exercise (B2), are a regression analogue of ranks: they depend on the data only through each observation's position relative to the fitted hyperplanes. Gutenbrunner and Jurečková (1992) showed that tests built from them are distribution-free in the sense classical rank tests are, and intervals follow by inverting the test over a grid. No density is estimated, which is why Koenker (2005, chapter 3) makes them the default; the cost is a grid search per interval.

The bootstrap. Resampling $(y_i, \mathbf{x}_{(i)})$ pairs — the case bootstrap of Definition 23.1.1(b) — needs only independence of the pairs, and Hahn (1995) proved that the bootstrap distribution of $\sqrt{n}(\hat{\beta}_\tau^* - \hat{\beta}_\tau)$ has the same limit as that of $\sqrt{n}(\hat{\beta}_\tau - \beta_\tau)$. Resampling *residuals* is not valid here, for the reason [Section 23.1](#) gave in the heteroscedastic linear model: it imposes the identical-distribution structure quantile regression was chosen to avoid.

Example. Which interval keeps its promise

Generate $n = 150$ observations from $y = 1 + x + (0.2 + 1.2x)\varepsilon$ with x uniform on $(0, 2)$ and ε standard normal, so the true slope of the τ -quantile is $1 + 1.2z_\tau$: 1.000 at the median, 2.538 at $\tau = 0.9$. Over 800 replicates (150 for the bootstrap, $B = 99$), the nominal 95% intervals cover:

method	$\tau = 0.5$	$\tau = 0.9$
iid formula	0.899	0.911
sandwich	0.965	0.911
pairs bootstrap	0.933	0.940

The iid formula undercovers at both levels, as it must. The sandwich is conservative at the median (mean width 0.989 against 0.743) and no better than the iid formula at $\tau = 0.9$, where its kernel estimate of $\mathbf{D}_1$ is fed by few residuals near zero. The bootstrap is the steadiest, at B extra linear programs.

45.3.5 Quantile curves that cross

Nothing in separate fitting forces the fitted curves to be ordered, yet the objects they estimate certainly are.

Proposition 45.3.3. Ordering, and how to restore it

- (a) Let $\bar{\mathbf{x}} = n^{-1} \sum_i \mathbf{x}_{(i)}$ and let $\hat{\boldsymbol{\beta}}_\tau$ solve Eq. 45.2.2. Then $\tau \mapsto \bar{\mathbf{x}}^\top \hat{\boldsymbol{\beta}}_\tau$ is nondecreasing: at the average design point the fitted quantiles never cross.
- (b) Fix $\mathbf{x}$ and levels $\tau_1 < \dots < \tau_m$, let $q_j = \mathbf{x}^\top \hat{\boldsymbol{\beta}}_{\tau_j}$ be the fitted values and let $q_{(1)} \leq \dots \leq q_{(m)}$ be the same numbers sorted. For any nondecreasing sequence $g_1 \leq \dots \leq g_m$ and any $p \geq 1$,

$$\sum_{j=1}^m |q_{(j)} - g_j|^p \leq \sum_{j=1}^m |q_j - g_j|^p. \quad (45.3.5)$$

Sorting the fitted quantiles therefore moves them no further from the true quantile curve, in every L^p sense at once.

Proof.

- (a) Split the check loss as $\rho_\tau(u) = \tau u + u_-$, where $u_- = (-u)_+$; this is an identity, since for $u \geq 0$ both sides are τu and for $u < 0$ both are $\tau u - u$. Summing over the residuals at $\mathbf{b}$,

$$R_\tau(\mathbf{b}) = \tau \sum_i (y_i - \mathbf{x}_{(i)}^\top \mathbf{b}) + S(\mathbf{b}) = \tau n(\bar{y} - \bar{\mathbf{x}}^\top \mathbf{b}) + S(\mathbf{b}), \quad S(\mathbf{b}) = \sum_i (\mathbf{x}_{(i)}^\top \mathbf{b} - y_i)_+.$$

Let $\tau_1 < \tau_2$ and abbreviate $\hat{\boldsymbol{\beta}}_j = \hat{\boldsymbol{\beta}}_{\tau_j}$. Optimality gives $R_{\tau_2}(\hat{\boldsymbol{\beta}}_2) \leq R_{\tau_2}(\hat{\boldsymbol{\beta}}_1)$ and $R_{\tau_1}(\hat{\boldsymbol{\beta}}_1) \leq R_{\tau_1}(\hat{\boldsymbol{\beta}}_2)$. Adding the two inequalities, the terms $S(\hat{\boldsymbol{\beta}}_1)$, $S(\hat{\boldsymbol{\beta}}_2)$ and $\bar{y}$ cancel and there remains

$$-\tau_2 \bar{\mathbf{x}}^\top \hat{\boldsymbol{\beta}}_2 - \tau_1 \bar{\mathbf{x}}^\top \hat{\boldsymbol{\beta}}_1 \leq -\tau_2 \bar{\mathbf{x}}^\top \hat{\boldsymbol{\beta}}_1 - \tau_1 \bar{\mathbf{x}}^\top \hat{\boldsymbol{\beta}}_2,$$

that is $(\tau_2 - \tau_1)(\bar{\mathbf{x}}^\top \hat{\boldsymbol{\beta}}_1 - \bar{\mathbf{x}}^\top \hat{\boldsymbol{\beta}}_2) \leq 0$. Since $\tau_2 > \tau_1$, $\bar{\mathbf{x}}^\top \hat{\boldsymbol{\beta}}_1 \leq \bar{\mathbf{x}}^\top \hat{\boldsymbol{\beta}}_2$.

- (b) Sorting a finite sequence is a finite composition of transpositions of adjacent inverted pairs, so it suffices to show that exchanging one inverted pair does not increase the sum. Suppose $i < j$ and $q_i > q_j$, and write $\phi(t) = |t|^p$, which is convex. Put $a = q_j - g_j$, $b = q_i - g_i$, $a' = q_i - g_j$, $b' = q_j - g_i$. Then $a' + b' = a + b$, and since $g_i \leq g_j$ and $q_j < q_i$ both a' and b' lie between $\min(a, b)$ and $\max(a, b)$. Two numbers with the same sum lying inside the interval spanned by another two can be written as $a' = \lambda a + (1 - \lambda)b$ and $b' = (1 - \lambda)a + \lambda b$ for some $\lambda \in [0, 1]$, so convexity gives $\phi(a') + \phi(b') \leq \phi(a) + \phi(b)$. Exchanging the pair replaces $\phi(b) + \phi(a)$ by $\phi(a') + \phi(b')$ and leaves every other term alone. ■

Part (b) is the **rearrangement** repair of Chernozhukov, Fernández-Val and Galichon (2010): fit the levels separately, then sort the fitted values at each $\mathbf{x}$. It costs nothing, can only help, and produces a genuine quantile function. Other repairs impose the ordering during fitting — He (1997) through a location-scale model, Bondell, Reich and Wang (2010) through linear constraints — at the price of a stronger model or a larger optimization.

Example. Where Engel's lines cross

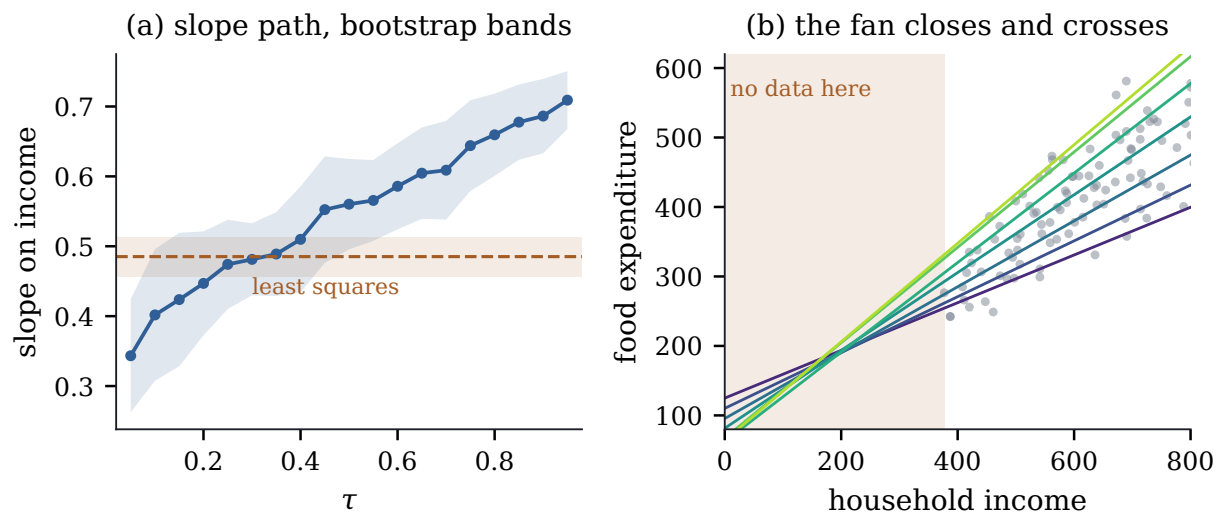


Figure 45.3.1: (a) The fitted income slope against τ , with pointwise 95% pairs-bootstrap bands and the least squares slope. (b) The seven fitted lines extended left of the smallest observed income: the fan closes and crosses.

The seven fitted lines of Example (Seven quantile lines for Engel's budgets) are not parallel, so every neighbouring pair crosses somewhere: at incomes of 252, 203, 163, 228, -117 and 143 francs. The smallest income in the data is 377, so every crossing lies outside the data, where the model was never tested (Figure 45.3.1(b)). Crossing reaches inside when the predictor is more flexible: adding a quadratic term in income leaves the curves out of order on 0.035 of a grid spanning the lowest 99% of incomes, beginning at the very edge. Rearranging repairs the ordering and moves the curves by at most 4.0 francs.

```
TAUS = (0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95)
beta = {tau: qreg(X, food, tau) for tau in TAUS}
crossings = {}
for lo, hi in zip(TAUS[:-1], TAUS[1:]):
    d0, d1 = beta[hi] - beta[lo]
    crossings[(lo, hi)] = -d0 / d1  # where two fitted lines meet
print(f"incomes in the data run from {income.min():.0f} to {income.max():.0f}")
print("neighbouring fitted lines meet at income",
      " ".join(f"{v:.0f}" for v in crossings.values()))
```

[Run this code online](#)

45.3.6 Two extensions, briefly

A Bayesian reading. Minimizing Eq. 45.2.2 maximizes the likelihood of an asymmetric Laplace density with location $\mathbf{x}_{(i)}^\top \boldsymbol{\beta}_\tau$ and asymmetry τ , so a flat prior makes the regression quantile a posterior mode and Chapter 39 (Definition 39.4.1, Definition 39.4.2) applies (Yu and Moyeed 2001; Kozumi and Kobayashi 2011 give a scale mixture yielding conditionally normal Gibbs steps). Only one τ can have the assumed error law, so the credible intervals deserve the scepticism due to any misspecified likelihood.

Nonlinear predictors. A roughness penalty on the check loss gives additive quantile regression — a linear program under an L^1 penalty (Koenker, Ng and Portnoy 1994), a quadratic program under the penalties of Chapter 43 — and structured additive predictors (Definition 44.5.1) carry over. The difficulty is the smoothing parameter: Theorem 29.3.2 needs a quadratic criterion, so

cross-validation must refit, or the problem must be handed to [Section 45.5](#).

45.3.7 Exercises

45.3.7.1 A. Check your understanding

Exercise. A1

Use [Eq. 45.3.3](#) to show that for iid normal errors the asymptotic variance of a regression median exceeds that of least squares by the factor $\pi/2$. Which assumption of [Theorem 45.3.2](#) is doing the work?

Solution. With $f(0) = 1/(\sigma\sqrt{2\pi})$ and $\tau = 1/2$, [Eq. 45.3.3](#) gives $(1/4)2\pi\sigma^2\mathbf{D}_0^{-1} = (\pi/2)\sigma^2\mathbf{D}_0^{-1}$, against $\sigma^2\mathbf{D}_0^{-1}$ for least squares. The identical-distribution assumption is what makes the comparison meaningful; under heteroscedasticity the comparison can go the other way, as [Example \(Three standard errors for one slope\)](#) shows in reverse.

45.3.7.2 B. Practice

Exercise. B1

For a standard normal error law, compute $\tau(1-\tau)s(\tau)^2$ at $\tau = 0.5, 0.9, 0.99$ and confirm the claim in the warning above that precision deteriorates in the tail. At which τ is the factor smallest?

Solution. $s(\tau) = 1/\varphi\{\Phi^{-1}(\tau)\}$, so the factor is $\tau(1-\tau)/\varphi\{\Phi^{-1}(\tau)\}^2$. At $\tau = 0.5$ it is $0.25/(0.3989)^2 \approx 1.571$; at $\tau = 0.9$, $0.09/(0.1755)^2 \approx 2.92$; at $\tau = 0.99$, $0.0099/(0.0267)^2 \approx 13.9$. Differentiating shows the minimum is at $\tau = 1/2$ by symmetry.

Exercise. B2

Two fitted quantile lines $a_1 + b_1x$ and $a_2 + b_2x$ with $b_1 < b_2$ cross at $x^* = -(a_2 - a_1)/(b_2 - b_1)$. Using the numbers of [Example \(Seven quantile lines for Engel's budgets\)](#), verify the crossing income of the $\tau = 0.05$ and $\tau = 0.10$ lines, and explain why [Proposition 45.3.3\(a\)](#) guarantees that x^* cannot lie at the mean income.

45.3.7.3 C. Going deeper

Exercise. C1

Use [Eq. 45.3.1](#) to show that the estimating equation for β_τ in the population is $E\{\mathbf{x}\psi_\tau(Y - \mathbf{x}^\top\beta_\tau)\} = \mathbf{0}$, and that this is the exact analogue of the normal equations with ψ_τ in place of the identity. Compare with the estimating-function framework of [Chapter 38](#) ([Definition 38.2.1](#)): is ψ_τ a quasi-score?